

ASSIGNMENT 2

INFOMDWR – Assignment 2: Data Integration & Preparation

Task 1: Profiling relational data

For this task, we used the file [Road Safety Data - Collisions - 2024](#) from the [road safety dataset](#) to calculate the summary statistics based on the discussion in the paper “Profiling Relational Data: A Survey” by Abedjan et al. (2015). We focus on the single-column profiling methods mentioned in it. We chose ten unique measures that provide a clean first glimpse at any set.

Selected statistics and their importance

1. Number of rows & columns – shows dataset size and complexity.
2. Data types – indicates whether attributes are numeric, text, or date/time.
3. Missing values – highlights completeness issues and cleaning needs.
4. Distinct values – shows attribute variability and possible keys.
5. Uniqueness ratio – quickly detects potential identifiers.
6. Mode (most frequent value) – shows common/default entries.
7. Constancy – reveals columns dominated by a single value.
8. Duplicate rows – identifies redundancy that can distort analysis.
9. Minimum & Maximum – reveal range and detect outliers.
10. Mean, Median, Standard Deviation – describe central tendency and spread; help understand skew and variability.

How we calculated them: we employed the pandas library since it has inbuilt methods for these statistics (shape, dtypes, isna().sum(), nunique(), mode(), duplicated(), min(), max(), mean(), median(), std()).

- We first imported the dataset with `pd.read_csv`.
- We then counted each measure individually and printed the outcome.
- For mode frequency (constancy) we applied `value_counts().max() / rows`.
- We chose only numeric columns for numeric summaries by selecting with `select_dtypes`.

This procedure follows the data profiling methodology in the paper: begin with column-level simple summaries in order to gain an understanding of structure, identify missing or repeated data, as well as likely issues with data quality prior to further preparation.

Task 2: Entity resolution

For this task, we used the [DBLP-ACM](#) dataset to find similar records using 2 methods: comparing all pairs of records and by using the LSH algorithm.

Task 2 - Part 1: Pairwise Comparison of All Records

In this part, we compare every single record in the dataset (ACM.csv) with all the records in (DBLP2.csv) to find the similar records (records that represent the same publication).

Before comparison, we start by loading the datasets and normalizing all records by converting all text to lowercase and converting multiple spaces to one.

We then wrote the functions to compute the following similarity measures:

- Levenshtein distance & similarity (for title)
- Jaro similarity (for author)
- Modified Affine gap similarity (for venue)
- Match (1) / Mismatch (0) (for year)

Using these similarity measures, we computed record similarity using the formula $\text{rec_sim} = w_1*s_t + w_2*s_a + w_3*s_c + w_4*s_y$ where $w_1+w_2+w_3+w_4 = 1$. Record pairs with $\text{rec_sim} > 0.7$ were classified as duplicates.

The pairwise comparison identified 2085 duplicate pairs which is a precision of 0.92 when validated against the ground truth mappings in DBLP-ACM_perfectMapping.csv. The running time of this method was about 50 mins.

The $O(n^2)$ time complexity of this approach makes it prohibitive. Parallelizing these comparisons would improve the running time without sacrificing accuracy since each comparison is independent. Other methods can be employed to avoid comparing all pairs of records – such as filtering on certain attributes (for example: year), or grouping by an attribute (like title) by its similarity measure first then only performing pairwise comparisons within that group. The latter method can affect accuracy though.

Task 2 - Part 2: Finding similar documents using LSH

In this part, we use the Locality Sensitive Hashing (LSH) algorithm to find similar records in the documents every single record in the [DBLP-ACM](#) dataset. We then compare the running time and precision of this method with that of part 1 above.

Again, we begin by loading the datasets and preparing the data. We concatenate the values in each record into one single string, change all alphabetical characters into

lowercase, and convert multiple spaces to one. Then we combine the records from both tables into one big list as we did during the lab.

Next we use the functions in the tutorials from lab 5 to compute the shingles, the minhash signature and the similarity on this list. We use $k=2$ since each document is fairly small, and `num_hashes = 1000` to create a highly accurate “fingerprint” for each record.

We then computed record similarity using the LSH algorithm and compare them to the actual mappings in the file `DBLP-ACM_perfectMapping.csv`. The precision of this method was 0.87 and the running time was 0.80 seconds.

Upon comparing with the precision and running time of part 1, we can see that running time of the LSH algorithm is over 3 orders of magnitude smaller than that of the pairwise comparison, whereas the precision is still fairly high.

Task 3: Data Preparation

For this task, we conducted data preparation steps on the [Pima Indians Diabetes Database](#). We began by removing the outcome column and then then computed the correlation between all pairs of remaining variables.

Then we removed the disguised values from the table, since values that equal to 0 from columns `BloodPressure`, `SkinThickness` and `BMI` as these are actually missing values but they have been replaced by the value 0. We replaced these 0 values with null for a more accurate representation of the data.

Next, we performed imputation by filling the cells with null using the mean values of the records that have the same class label. We again computed the correlation between all pairs of variables and compared these values with the correlation values obtained in the first step.

Upon comparison, we noted the following:

- The correlation b/w bloodpressure and pregnancies increased from 0.14 to 0.20 appx. once the mean values were replaced with null.
- skinthickness and pregnancies showed a negative corr of 0.08. However, replacing with mean resulted in a positive corr of 0.09. Since the correlation is rather weak, no relationship amongst these variables can be established based on these results.
- Glucose positive correlation with bloodpressure increased to 0.22 from 0.15, with skinthickness a significant increase to 0.22 from 0.05.
- Skinthinness and insulin showed a significant decrease in correlation, from 0.43 to 0.10, the result puts doubt on the relation between these variables.

Conversely, with BMI the correlation increased, this aligns with the natural phenomenon of fat accumulation with high BMI and thickness in the skin. Similarly the correlation with age went from negative to positive aligning with what's naturally observed.